

MKTG100 Numeracy

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Module Outline

Preface

Employers such as firms and government agencies expect marketing graduates to have good analytical thinking and basic numeracy skills. However, many of you might not have a high confidence when it comes to dealing with numbers. Anecdotal evidence shows that majority of marketing students choose the subject because “...they perceive a lower-level requirement of quantitative skills in marketing as compared to other business disciplines such as accounting, finance, and information technology” (Ganesh, Sun, and Barat 2010, 47). This is in contrast to the high demand and expectation from employers for these skills for marketing graduates. We want to help you to meet the challenge and design this course. In return, we want you to be seriously engaged in this module.

The Team

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Tutor:

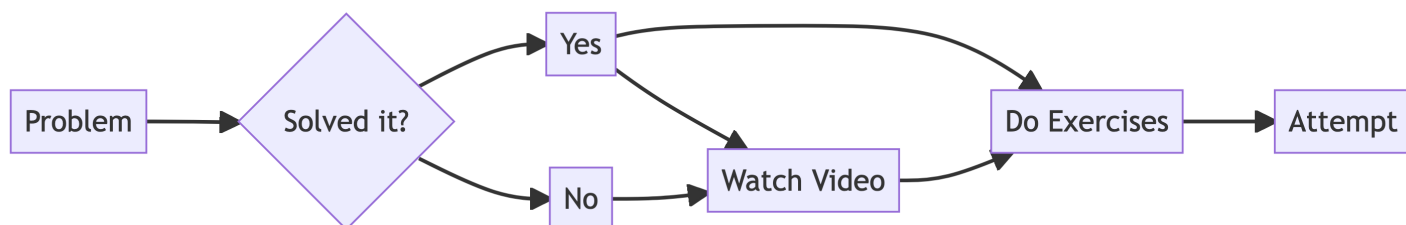
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Learning Objectives

1. To improve basic numeracy skills.
2. To increase students' appreciation of the importance of gaining those skills in the marketing environment.
3. As a preparation for advanced modules in Part 2.

How to Study

Each section in this website relates to a particular topic. The section has a skeleton that includes **Problem**, **Watch**, and **Exercises**. Your learning journey on a topic will take the following routes described by this flowchart:



Program Structure

The module runs from week 5 - 9 in the Michaelmas term. In week 9, you take a test on Moodle during your workshop.

All workshop activities are conducted online. You must log onto **teams**. Your tutors are available virtually during workshop time. If you need any help, you can send a message or request to talk to your tutors.

During the workshops, you must attempt the weekly exercises. If you cannot finish them during the workshop, you must complete them outside of the workshop time and make sure they are finished before the next session the week after.

The structure of the program can be seen below.

Week	Section	Topic
5	5	Arithmetic of whole numbers
6	6	Fraction and Decimal Numbers
7	7a	Percentage and Ratio
7	7b	Exponents and Roots
8	8a	Basic Marketing Math: Business Costs and Break-Even
8	8b	
9	All	Exam (on Moodle)

Exercises and Week 9 Test on Moodle

At the end of your journey (see flowchart), you must attempt the weekly exercises on Moodle. Direct links to the weekly exercises are given at the end of each page or alternatively you can log onto Moodle and visit the course page. Once you submit all your answers, you have completed the weekly topic.



Figure 1: Weekly Exercises and Week 9 Test On Moodle

Assessment

Exam or test will take place on Moodle in the last workshop of week 9. The exam will last for 40 min for 20 questions. The pass mark is 40%. If you fail, you will have a chance to take a resit – the fails will be contacted by a module leader. The test link on Moodle for week 9 test is **password protected**. Your tutor will reveal the password in the workshop.

Make sure you don't miss out your examinations as they are mandatory for completion of your module.

Textbook

A textbook is not needed. The information in this site is meant to be self-sufficient. But if you really want to buy a book, I recommend Daryanto (2020). This is a compilation book whereby its chapters are taken from Croft, Croft, and Davison (2006) and Berenson et al. (2012). The book had been used as a textbook of the module in previous years. It also contains chapters on basic statistics covered in MKTG200 Basic Statistics, which you will take in your second year. *Several copies of the book are available in the library.*

Excel

I assume you know how to use **Excel** to calculate a basic mathematical operation e.g., $1+1$. To answer certain problems in this manual, I provide **Excel** commands. This is a nice tool to use but also dangerous. Without an understanding, the use of **Excel** can put you in the dark forever. Make sure you understand how to manually solve the weekly problems/exercises before resorting to **Excel**.

1 Week 5: Arithmetic

1.1 Learning Goals

- Understand what arithmetic is
- Understand BODMAS rule
- Understand Prime numbers
- Know to how do prime factorisation
- Understand highest common factors (hcf) / greatest common divisor (gcd)
- Understand lowest common factors (lcm)

1.2 What Is Arithmetic?

Arithmetic is a branch of mathematics that deals with numbers and their operations, which includes addition, subtraction, multiplication, and division.

1.2.1 Problem

What is the answer to a multiplication problem called?

Select the correct answer:

- a. Product
- b. Sum
- c. Quotient
- d. Difference

1.2.2 Watch

[Click this video](#)

1.2.3 Exercises

1. What is the answer to a multiplication problem called?
2. What is the answer to a division problem called?
3. What is the answer to an addition problem called?
4. What is the answer to a subtraction problem called?

1.3 Order of Operations: BODMAS rule

“Operations” refer to mathematical operations such as addition, subtraction, multiplication, division, taking power (a^n) or roots ($\sqrt[n]{a}$) (note: $\sqrt[n]{a} = a^{1/n}$).

1.3.1 Problem

How to work this out: $10:2(2+1) = ?$

What to do first? The answer is BODMAS.

BODMAS = **B**rain, **O**rders, **D**ivision, **M**ultiplication, **A**ddition, and **S**ubtraction.

Table 1.1: Order of Operations and Priority

Symbol	Meaning	Operation	Priority
B	Bracket	$()$	1
O	Order	a^n	1
D	Division	$\div$	2
M	Multiplication	$\times$	2
A	Addition	$+$	3
S	Subtraction	$-$	3

! Important

You work out an expression from left to right in the order of BODMAS. Division and Multiplication are interchangeable (i.e., they have the same priority) and depend on which comes first in the expression. Similar rule applies for addition and subtraction as they have the same priority.

1.3.2 Watch

[Click this video](#)

i Millions people got it wrong: $6:2(1+2) = ?$

A BODMAS problem $6:2(1+2)$ got viral on the internet. Read the news here
<https://www.newsweek.com/this-viral-math-problem-has-internet-frenzy-1584710>
Watch the explanation in the video below.
[Click this video](#)

1.3.3 Exercises

Note: * symbol is the same as $\times$

1. $3 + 4 * 2 = \dots$
2. $(1 + 2) 5 - 2 * 3 = \dots$
3. $8 : 2 (2 + 1) = \dots$
4. $2 - 2 + 2 = \dots$
5. $(10 : 2) * 3 - 2 = \dots$
6. $5 * 5^2 = \dots$

1.4 Prime Numbers

1.4.1 Problem

Is 5 a prime number?

1.4.2 Watch

[Click this video](#)

1.4.3 Exercises

1. Is 13 a prime number?
2. Is 33 a prime number?

1.5 Prime Factorization

1.5.1 Problem

Find the prime factorization of 75.

1.5.2 Watch

[Click this video](#)

1.5.3 Exercises

1. Find the prime factorization of 136.
2. Find the prime factorization of 100.
3. Find the prime factorization of 124.

1.6 Greatest common factors (gcd)

The greatest common divisor (gcd) is also called the highest common factor (hcf). GCD of two numbers is the number that divides them both. For instance GCD of 3 and 6 is 3.

1.6.1 Problem

What is the greatest common divisor of 18 and 21?

1.6.2 Watch

[Click this video](#)

1.6.3 Exercises

1. Find the hcf of 12 and 20.
2. Find the gcd of 35 and 100.
3. Find the gcd of 3 and 12.

1.6.4 Check Your Calculation

 Excel

Type in:
= GCD(3,6,12)
press **Enter**

1.7 Lowest common factors (lcm)

The lowest common factors is also called the the lowest common multiple. LCM of number A (say 2) and B (say 5) is the lowest number that can be divided by A and B without a remainder. For example, the lcm for 2 and 5 is 10 because 10 can be divided by 2 and 5 without a remainder.

1.7.1 Problem

Find the lcm of 6 and 10.

1.7.2 Watch

[Click this video](#)

1.7.3 Exercises

1. Find the lcm of 2 and 5.
2. Find the lcm of 3 and 6.

1.7.4 Check Your Calculation

 Excel

Type in
= LCM (3,6)
press **Enter**

1.8 Attempt the Exercises on Moodle

At this point, you would have worked on the above exercise questions.

Now, I want you to attempt these questions on Moodle.

Please click the link below:

[Week 5 Exercises on Moodle](#)

Once you submit all your answers on Moodle, you have completed your learning on this week's topic.

2 Week 6: Fraction, Decimals, and Significant Figures

2.1 Learning Goals

- Understand fraction and decimal number
- Understand rounding a number
- Understand significant figures

2.2 Fraction and Decimal Number

Fraction

$\frac{1}{2}$ and $\frac{4}{3}$ are fractions. The top number in a fraction is called **numerator**, and the bottom number is called **denominator**. $\frac{1}{2}$ is a proper fraction.

! Important

A proper fraction is a fraction in which the numerator is less than the denominator.

$\frac{4}{3}$ is not a proper fraction. We can write $\frac{4}{3}$ as $1\frac{1}{3}$. The latter is a mixed fraction. A mixed fraction is a mix of a whole number and a proper fraction.

$\frac{2}{4}$ is a proper fraction because its numerator is less than the denominator. However, $\frac{2}{4}$ is not in its simplest form because it can be simplified as $\frac{1}{2}$.

Decimal

Decimal means a number based on 10. The word comes from Latin decima: a tenth part. 56.8 has 5 tens, 6 ones and 8 tenths.

⚠ Warning

In the UK system, the decimal point is denoted by a **dot**. In other countries, the decimal point is denoted by a **comma**!

2.2.1 Problem

Express 0.142 as a proper fraction in its simplest form.

2.2.2 Watch

[Click this video](#)

[Click this video](#)

[Click this video](#)

[Click this video](#)

2.2.3 Exercises

1. Express 0.015 as a proper fraction in its simplest form.
2. Express 4.6 as a mixed fraction in its simplest form.
3. Write $\frac{1}{100} + \frac{5}{1000}$ as a decimal number.
4. Round 24.362 to 1 decimal places (1 dp).
5. Round 0.0452 to 2 dp.
6. In this fraction $\frac{1}{6}$, the numerator is ... and the denominator is

2.3 Significant Figures

Significant means important. If I want you to report a number with 2 significant figures, it means the two digits in the number are important.

Significant figures of a number relates to how we want to express the accuracy of the number. To decide how many significant figures we want to report, we count at the first non-zero digit. All the zeros **after the decimal point** that are on the right of the last non-zero digit are considered to be significant. For example, 0.05670 contains four significant digits; 8150 contains four significant figures.

2.3.1 Problem

Write 76 to one significant figure (1 sf).

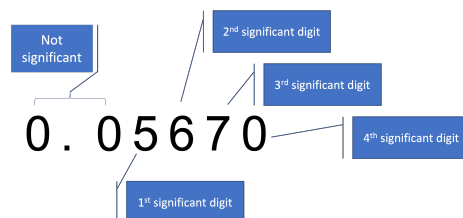


Figure 2.1: 0.05670 contains four significant figures

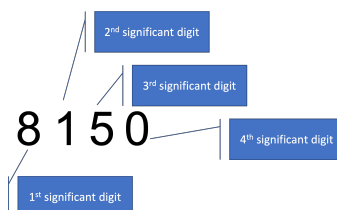


Figure 2.2: 8150 contains four significant figures

2.3.2 Watch

[Click this video](#)

2.3.3 Exercises

1. Write 52 to three significant figure (3 sf).
2. Write 35.572 to 3 sf.
3. Write 0.0473 to 1 sf.
4. Write 695.3 to 2 sf.

2.4 Attempt the Exercises on Moodle

At this point, you would have worked on the above exercise questions.

Now, I want you to attempt these questions on Moodle.

Please click the link below:

[Week 6 Exercises on Moodle](#)

Once you submit all your answers on Moodle, you complete your learning on this week's topic.

3 Week 7a: Percentages and Ratio

3.1 Learning goals

- Understand the meaning of percentages and ratio.
- Know how to perform calculation with percentages and ratio.
- Know how to calculate percentage changes.

3.2 Percentages and Ratio

3.2.1 Problem

Transform 3.6 into a percentage.

3.2.2 Watch

[Click this video](#)

3.2.3 Exercises

1. Transform 2.5 into a percentage.
2. Calculate 12% of 60.
3. Calculate 11.4% of 100.

3.3 Calculating Percentages Changes

$$\begin{aligned}\text{Percentages changes} &= \frac{\text{Change between values}}{\text{Original value}} \times 100\% \\ &= \frac{\text{New value} - \text{Original value}}{\text{Original value}} \times 100\%\end{aligned}$$

- If the percentage change is positive, it is called a percentage increase.
- If the percentage change is negative, it is called a percentage decrease.

Calculating new price.

This year, your monthly rent is £750. Next year, the monthly rent is increased by 4%. Calculate the new rent.

Answer: The new rent is $(100\% + 4\%) \times £750 = 104\% \times £750 = £780$

3.3.1 Problem

- Last month, 250 consumers bought a new smartphone. This month, the number is 420. What is the percentage increase in sale?
- This year, your monthly rent is £750. Next year, the monthly rent is reduced by 5%. Calculate the new rent.

3.3.2 Watch

[Click this video](#)

3.3.3 Exercises

1. Last month, 250 consumers bought a new smartphone. This month, the number is down to 100. What is the percentage decrease in sale?
2. Yesterday there were 250 unique visitors clicking a website. Today, you found that there is a 10% increase in visit. How many clicks are there today?

3.4 Attempt the Exercises on Moodle

At this point, you would have worked on the above exercise questions.

Now, I want you to attempt these questions on Moodle.

Please click the link below:

[Week 7a Exercises on Moodle](#)

Once you submit all your answers on Moodle, you complete your learning on this week's topic.

4 Week 7b: Exponents and Roots

4.1 Learning Goals

- Know how to evaluate exponential expressions
- Know how to write exponential expressions when multiplying or dividing with the same base.
- Know how to find roots.

4.2 Exponents

Rules:

1. $a^m \times a^n = a^{m+n}$
2. $\frac{a^m}{a^n} = a^{m-n}$
3. $(a^m)^n = a^{mn}$
4. $a^{-m} = \frac{1}{a^m}$

4.2.1 Problems

Simply:

1. $x^2 \times x^3 = \dots$
2. $\frac{3^5}{3^2} = \dots$
3. $(4^5)^2 = \dots$
4. $2^{-3} = \dots$

4.2.2 Watch

[Click this video](#)

4.2.3 Exercises

Simply:

1. $z^3 \times z^2 = \dots$
2. $\frac{4^3}{4^2} = \dots$
3. $(5^2)^3 = \dots$
4. $3^{-2} = \dots$

4.3 Attempt the Exercises on Moodle

At this point, you would have worked on the above exercise questions.

Now, I want you to attempt these questions on Moodle.

Please click the link below:

[Week 7a Exercises on Moodle](#)

Once you submit all your answers on Moodle, you complete your learning on this week's topic.

4.4 Roots

Square roots. 4 is called a square root of 16 because $4^2 = 16$. The symbol $\sqrt[2]{}$ or $\sqrt{}$ is used to represent square root. For example, $\sqrt{16} = 4$. $\sqrt{16}$ can be written as $16^{\frac{1}{2}}$.

Cube roots. 2 is called a cube root of 8 because $2^3 = 8$. The symbol $\sqrt[3]{}$ is used to represent cube root. For example, $\sqrt[3]{8} = 2$. $\sqrt[3]{8}$ can be written as $8^{\frac{1}{3}}$.

** In general, $\sqrt[n]{a}$ can be written as $a^{\frac{1}{n}}$. It is hard sometimes to evaluate this expression manually. You can use a scientific calculator or **Excel** to do it. For example, by typing `=16^(1/4)` in **Excel**, it gives 2. That is $\sqrt[4]{16} = 16^{\frac{1}{4}} = 2$. You can prove that $2 \times 2 \times 2 \times 2 = 16$

4.4.1 Problems

Evaluate $\sqrt[3]{16}$ (you can use a scientific calculator or **Excel**)

4.4.2 Watch

[Click this video](#)

4.4.3 Exercises

Evaluate

1. $\sqrt[3]{81}$
2. $\sqrt[4]{256}$
3. $\sqrt{81}$
4. $\sqrt[3]{216}$
5. $\sqrt[4]{625}$
6. $\sqrt[3]{15.63}$

4.5 Attempt the Exercises on Moodle

At this point, you would have worked on the above exercise questions.

Now, I want you to attempt these questions on Moodle.

Please click the link below:

[Week 7b Exercises on Moodle](#)

Once you submit all your answers on Moodle, you complete your learning on this week's topic.

5 Week 8a: Business Cost and Break-Even Analysis

5.1 Learning Goals

Know how to calculate

- Fixed cost and variable cost
- Break-even analysis

5.2 Fixed cost and variable cost

Fixed costs are expenses that a firm must pay whatever amounts of products the firm sells. Examples of fixed cost are rent and advertising expenditure.

Variable costs are expenses that varies in relation to how many products the firm sells. Examples of variable costs are delivery costs, packaging costs.

Total cost = Fixed Costs + Variable Costs $\times$ Outputs

5.2.1 Problem

The cost of making a one unit of a product is £40. What is the cost of making 20 units?

5.2.2 Watch

[Click this video](#)

5.2.3 Exercises

Look at the table below.

Calculate total cost.

Your Business (December)	
Item	Cost (£)
Variable costs per job	£50.00
Office rent	£1,500.00
Wages	£5,000.00
Advertising	£2,000.00
Other fixed costs	£800.00
Expected number of jobs per month	50

5.3 Break-even analysis

$$\text{Break – even quantity} = \frac{\text{Fixed cost}}{\text{Contribution}}$$

$$\text{Contribution} = \text{Selling Price} - \text{Variable Cost}$$

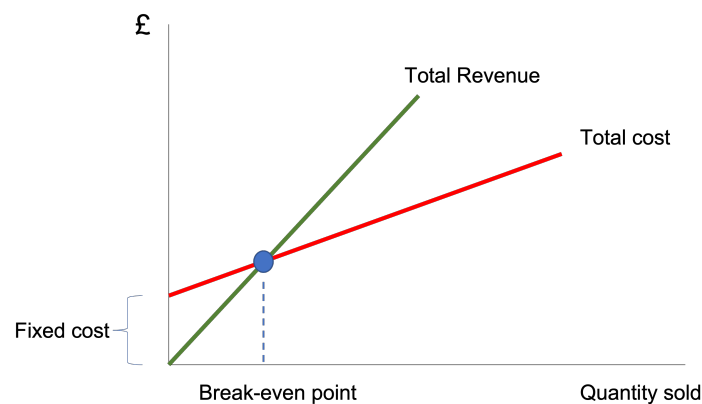


Figure 5.1: Break-even chart

💡 Tip

When you calculate the break-even quantity, the number can be with decimals (e.g., 25.3 units). How would you handle the decimal? You can not sell 25.3 units. What you can do is to round the number up to the next whole number regardless of the decimals. In this example, the next whole number is 26. Thus, you must sell 26 units.

5.3.1 Problem

You sell a product with a price of £20 per unit. The variable cost of the product is £10 per unit. The fixed costs associated with the product is £20,000. How many products should you

sell to achieve break-even?

5.3.2 Watch

[Click this video](#)

5.3.3 Exercises

1. A company's variable costs per product are £5. The daily fixed costs are £100. If the product is sold at £10, calculate the break-even quantity.
2. If you increase the selling price by 5%, calculate the new break-even quantity.

5.4 Attempt the Exercises on Moodle

At this point, you would have worked on the above exercise questions.

Now, I want you to attempt these questions on Moodle.

Please click the link below:

[Week 8a Exercises on Moodle](#)

Once you submit all your answers on Moodle, you complete your learning on this week's topic.

6 Week 8b: Market Share

6.1 Learning Goals

- Understand market share

6.2 Market Share

Market share of a firm is the firm's market share of the served market. Market share is one of the determinants how profitable a business is. A firm that has a high market share is more likely to be profitable than their smaller-share competitors. Market share can be calculated based on sales: the percentage of firm's sales obtained by the firm:

$$\text{Market Share (\%)} = (\text{Firm's Sales} / \text{Total Market Sales}) \times 100$$

6.2.1 Problem

Firm A and Firm B are two companies competing in the luxury car market. Firm A sold 330k units last year, while Firm B sold 440k units in the same period. Calculate the market share of each company based on the given sales figures.

6.2.2 Watch

[Click this video](#)

6.2.3 Exercises

1. In the leather jacket market, two companies are the main players in the market that fiercely compete to each other. Company X reported GBP 120 million in sales last year, while Company Y reported GBP 80 million. Calculate the market share based on the sales revenue.

Now, imagine that Company X launches a new marketing campaign that increases its sales to \$140 million this year, while Company Y's sales remain the same. Calculate the new market shares for both companies.

2. Read this statement: "In June 2023, Facebook's market share reached 55 percent in the United Kingdom..." (www.statista.com)

Which of the following is the correct interpretation of the above statement?

- A) The percentage of UK's population using Facebook in June 2023.
- B) The percentage of Facebook's global users located in the United Kingdom.
- C) The percentage of total advertising revenue generated by Facebook in the UK.
- D) The percentage of UK's internet users using Facebook compared to other platforms.

6.3 Attempt the Exercises on Moodle

At this point, you would have worked on the above exercise questions.

Now, I want you to attempt these questions on Moodle.

Please click the link below:

[Week 8b Exercises on Moodle](#)

Once you submit all your answers on Moodle, you have completed your learning on this week's topic.

7 Week 9 Test

This is an exam week for this module. The exam will last for 40 min on Moodle. There are 20 questions of 20 points in total. The pass mark is 8 points (40%). The pass mark is quite low given the level of the questions. Therefore, I don't expect to see fails on this test unless you have never attempted any of the weekly problems/exercises. Furthermore the test contain questions of the weekly test questions.

! Important

Make sure you don't miss out your examinations as they are mandatory for the completion of your module.

The link on Moodle for week 9 test is **password protected**. Your tutor will reveal the password for you in your workshop.

Now, go to Moodle, and click the link 'Week 9 Test'

Alternatively, if you are connected to the internet campus, you can directly click this link [Week 9 Test](#)

References

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